FIT3155

Advanced Algorithms and Data Structures

Week 4 Example: Linear Time Suffix Tree Construction

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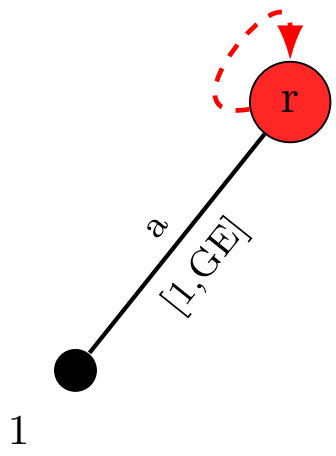
Phase 1 Extension 1

ext phase

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
a	b	b	b	b	c	b	b	c	b	c	a	b	b	b	b	$



Phase 1, Extension 1

i : 1
j : 1
GE: 1

Before Extension 1

AN: r
rem: ‘ ’

Action

Skip count: Not applicable
Apply **Rule 2 Alt**
Leaf node created (1)

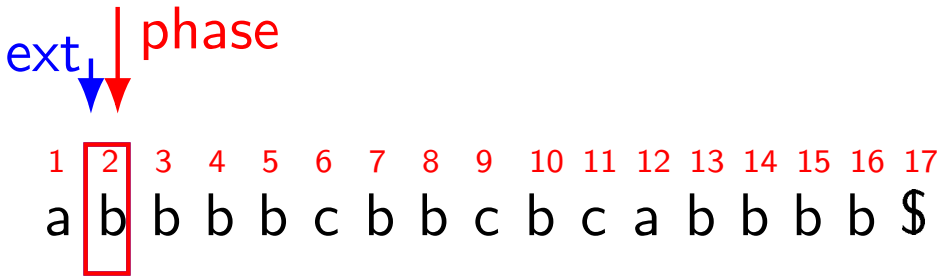
After Extension 1

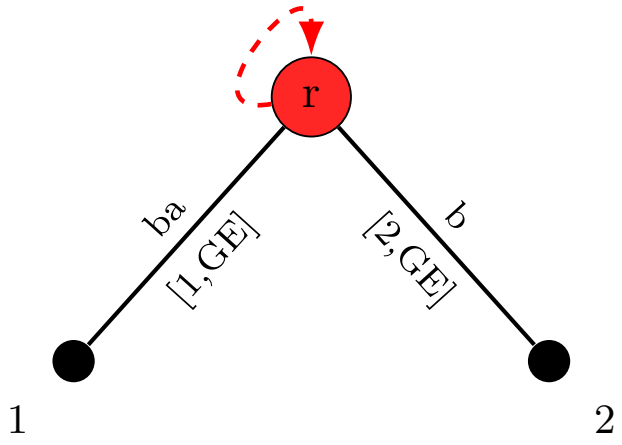
AN : r (fllowing link r → r)
rem: ‘ ’

Suffix links

Pending: None
Resolved: None

Phase 2 Extension 2





Phase 2, Extension 2

i :	2
j :	2
GE:	2

Before Extension 2

AN:	r
rem:	‘ ’

Action

Skip count: Not applicable
Apply **Rule 2 Alt**
Leaf node created (2)

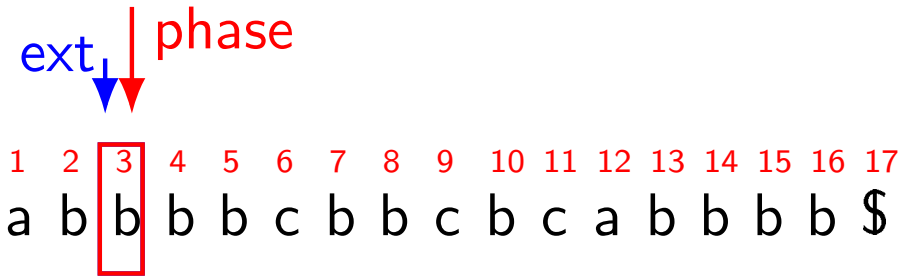
After Extension 2

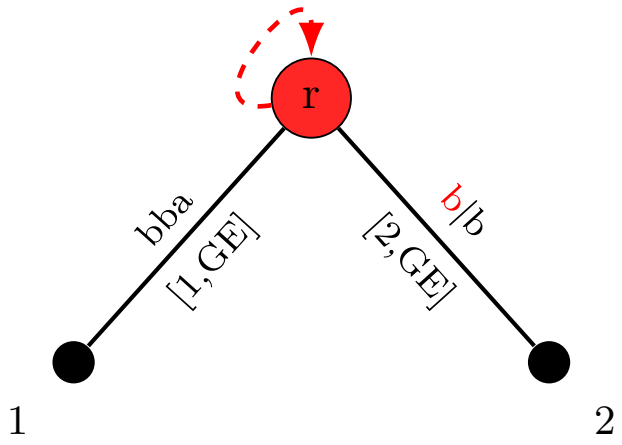
AN:	r (fllowing link r $\rightarrow$ r)
rem:	‘ ’

Suffix links

Pending: None
Resolved: None

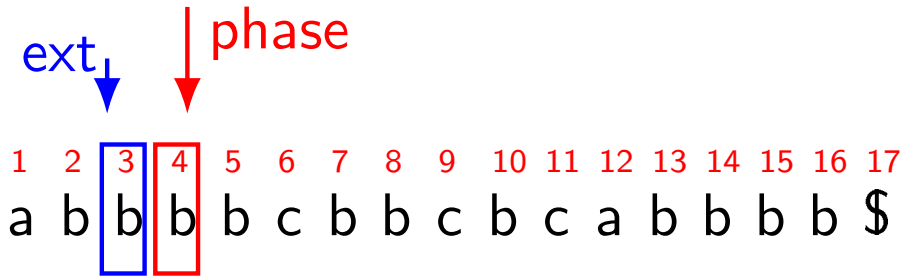
Phase 3 Extension 3

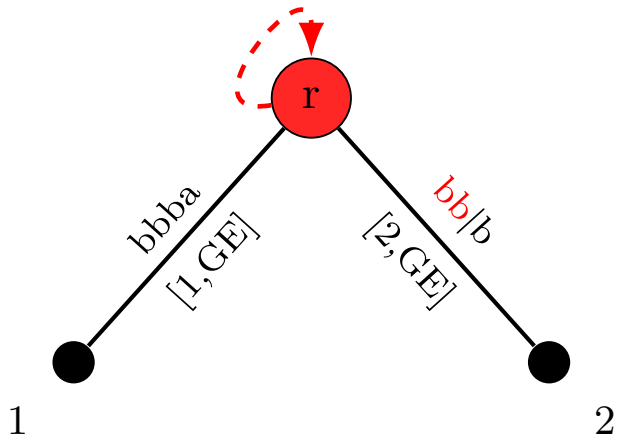




Phase 3, Extension 3	
i :	3
j :	3
GE:	3
Before Extension 3	
AN:	r
rem:	‘ ’
Action	
Skip count:	Not applicable
Apply **Rule 3**	
After Extension 3	
AN :	r (As previous extension)
rem:	‘b’
Suffix links	
Pending :	None
Resolved:	None

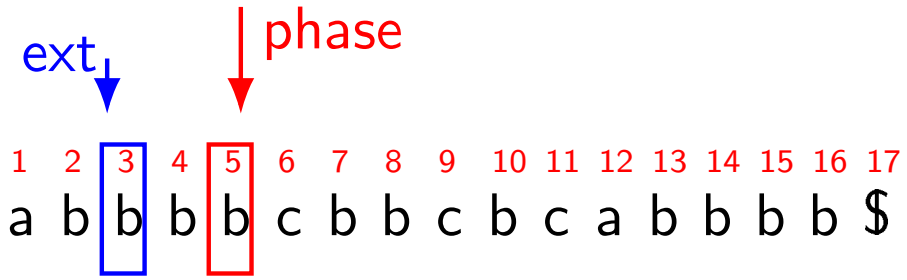
Phase 4 Extension 3

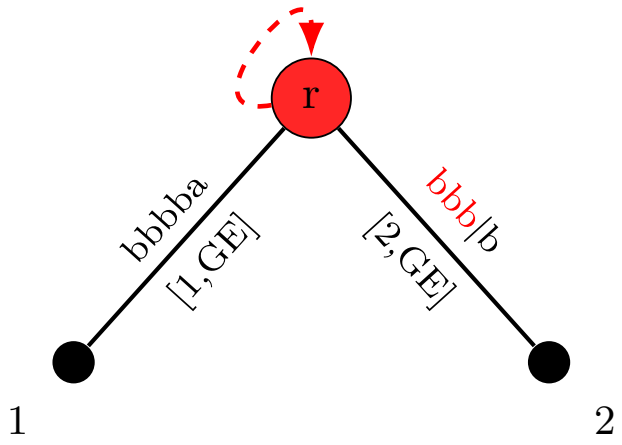




Phase 4, Extension 3	
i :	4
j :	3
GE:	4
Before Extension 3	
AN :	r
rem:	'b'
Action	
Skip count: 'b' from AN=r	
Apply **Rule 3**	
After Extension 3	
AN :	r (As previous extension)
rem:	'bb'
Suffix links	
Pending :	None
Resolved:	None

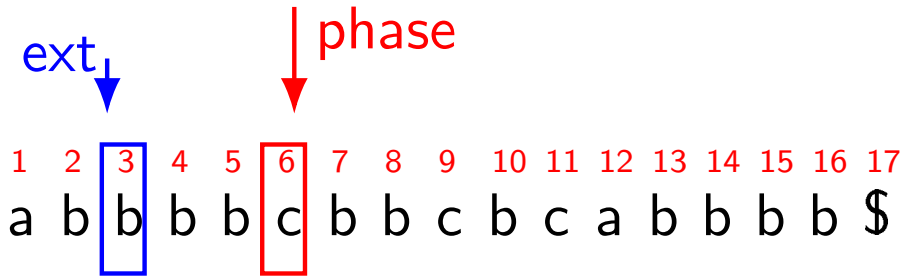
Phase 5 Extension 3

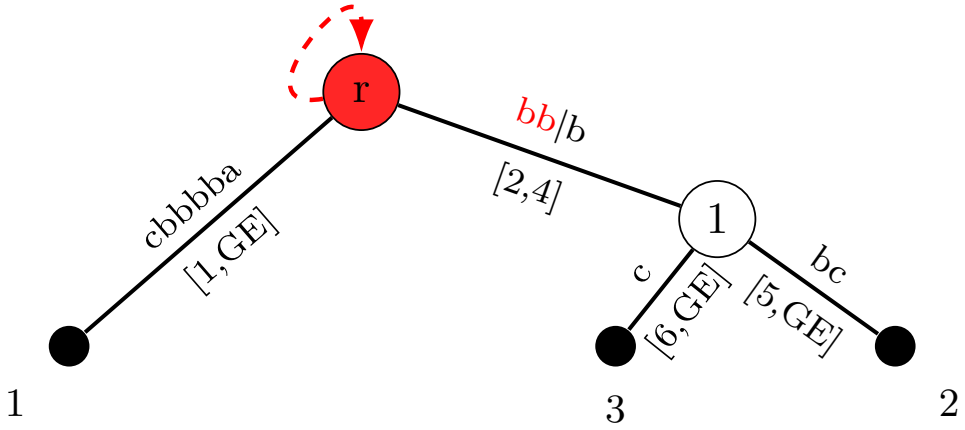




Phase 5, Extension 3	
i :	5
j :	3
GE:	5
Before Extension 3	
AN :	r
rem:	'bb'
Action	
Skip count: skip 'bb' from r	
Apply **Rule 3**	
After Extension 3	
AN :	r (As previous extension)
rem:	'bbb'
Suffix links	
Pending :	None
Resolved:	None

Phase 6 Extension 3





Phase 6, Extension 3

i :	6
j :	3
GE:	6

Before Extension 3

AN :	r
rem:	'bbb'

Action

Skip count: skip 'bbb' from r
Apply **Rule 2 Reg**
Internal node created (1)

After Extension 3

AN :	r (fllowing link r $\rightarrow$ r)
rem:	'bb' (remove 1st char)

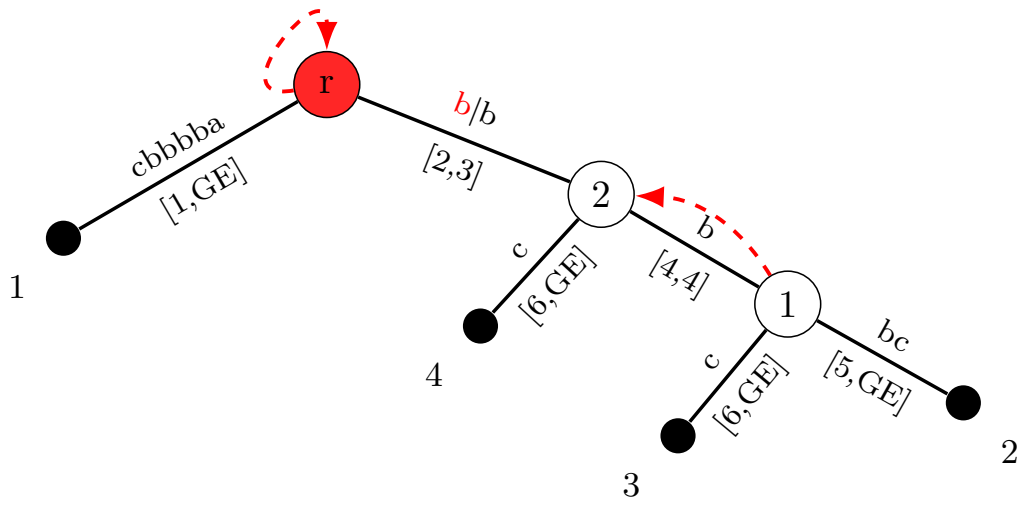
Suffix links

Pending : Node 1
Resolved: None

Phase 6 Extension 4

ext phase

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
a	b	b	b	b	c	b	b	c	b	c	a	b	b	b	b	$



Phase 6, Extension 4

i : 6
j : 4
GE: 6

Before Extension 4

AN : r
rem: 'bb'

Action

Skip count: skip 'bb' from r
Apply **Rule 2 Reg**
Internal node created (2)

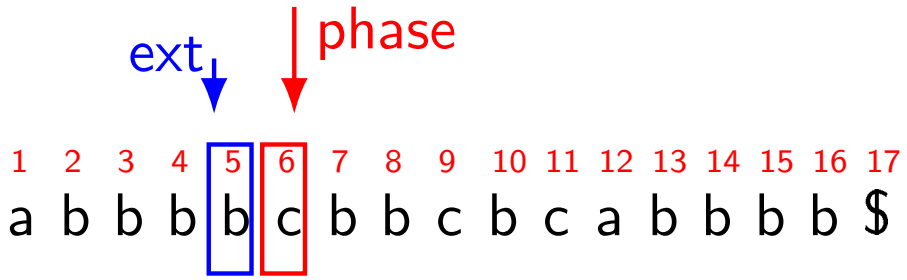
After Extension 4

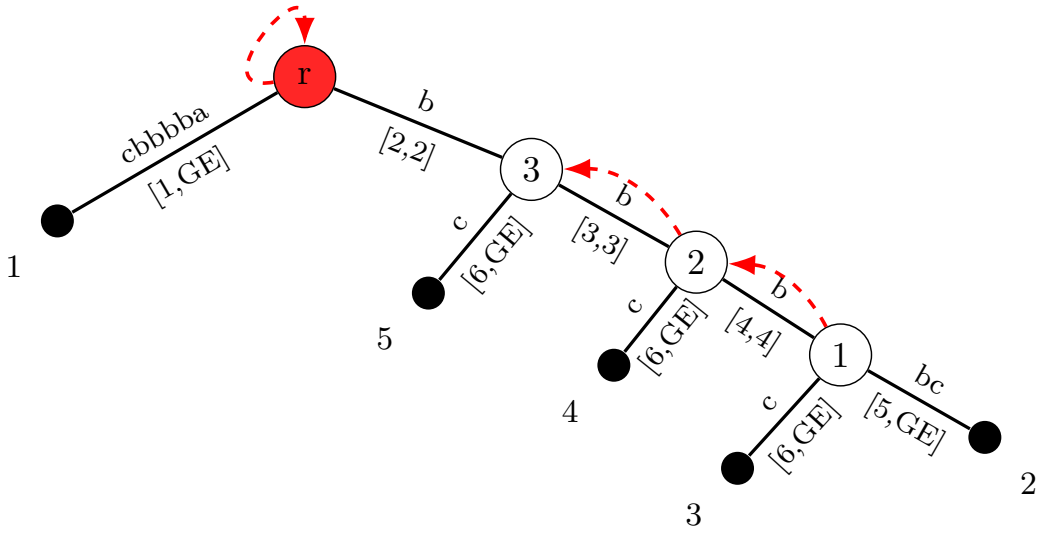
AN : r (fllowing link r → r)
rem: 'b' (remove 1st char)

Suffix links

Pending : node 2
Resolved: 1 → 2

Phase 6 Extension 5





Phase 6, Extension 5

i :	6
j :	5
GE:	6

Before Extension 5

AN :	r
rem:	'b'

Action

Skip count: skip 'b' from AN
Apply **Rule 2 Reg**
Internal node created (3)

After Extension 5

AN :	r (fllowing link r → r)
rem:	' ' (remove 1st char)

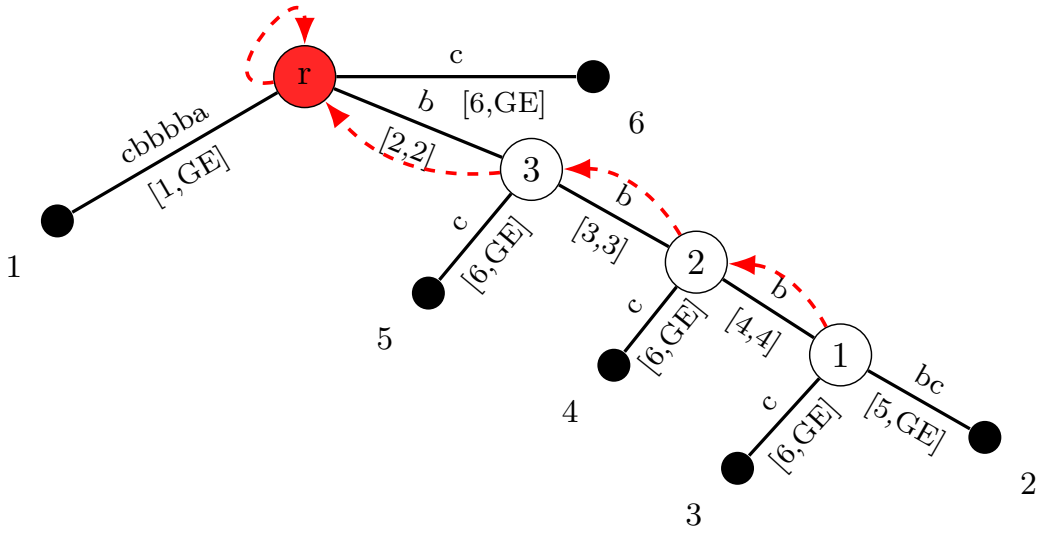
Suffix links

Pending : Node 3
Resolved: 2 → 3

Phase 6 Extension 6

ext phase

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
a	b	b	b	b	c	b	b	c	b	c	a	b	b	b	b	$



Phase 6, Extension 6

i : 6
j : 6
GE: 6

Before Extension 6

AN : r
rem: ‘ ’

Action

Skip count: skip ‘b’ from AN
Apply **Rule 2 Alt**
Leaf node created (6)

After Extension 6

AN : r (fllowing link r → r)
rem: ‘ ’

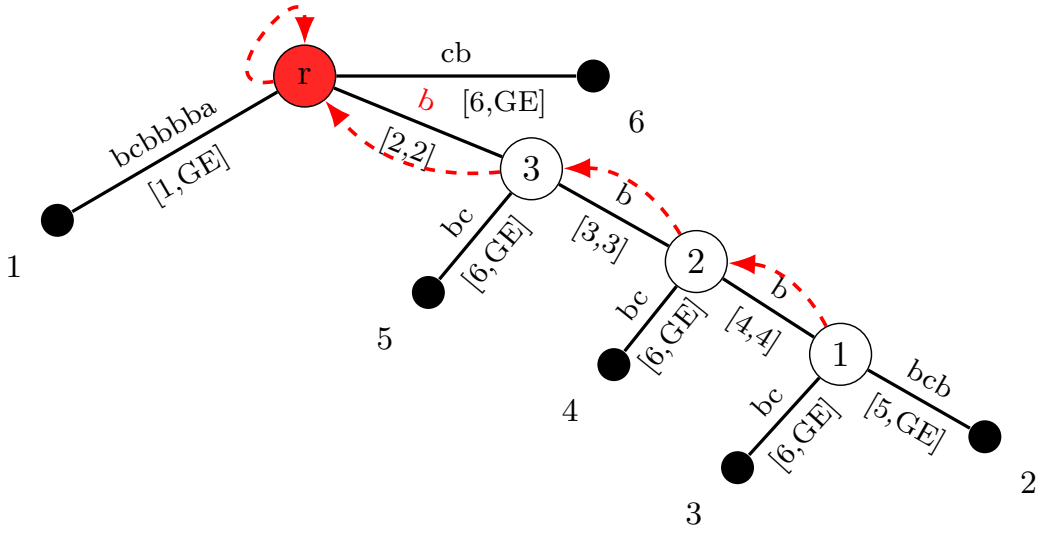
Suffix links

Pending : None
Resolved: 3 → AN=r

Phase 7 Extension 7

ext phase

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
a	b	b	b	b	c	b	b	c	b	c	a	b	b	b	b	$



Phase 7, Extension 7

i : 7
j : 7
GE: 7

Before Extension 7

AN : r
rem: ‘ ’

Action

Skip count: Not applicable
Apply **Rule 3**

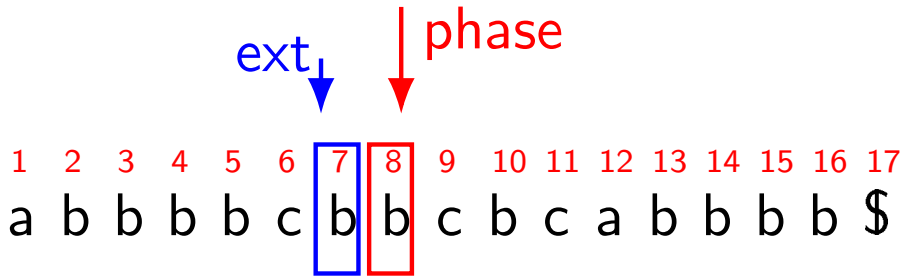
After Extension 7

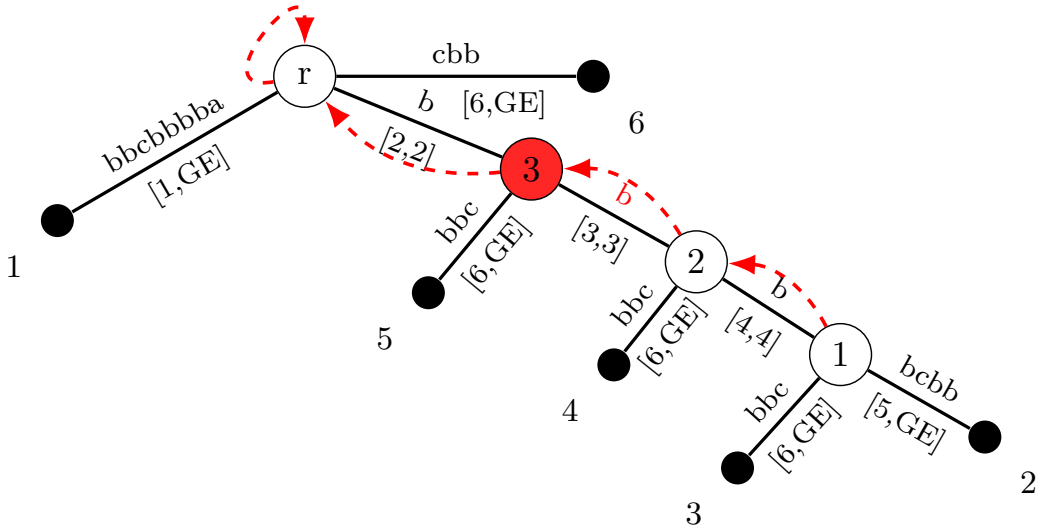
AN : r (As previous extension)
rem: ‘b’

Suffix links

Pending : None
Resolved: None

Phase 8 Extension 7





Phase 8, Extension 7

i :	8
j :	7
GE:	8

Before Extension 7

AN :	r
rem:	'b'

Action

Skip count: skip 'b' from AN

Apply **Rule 3**

After Extension 7

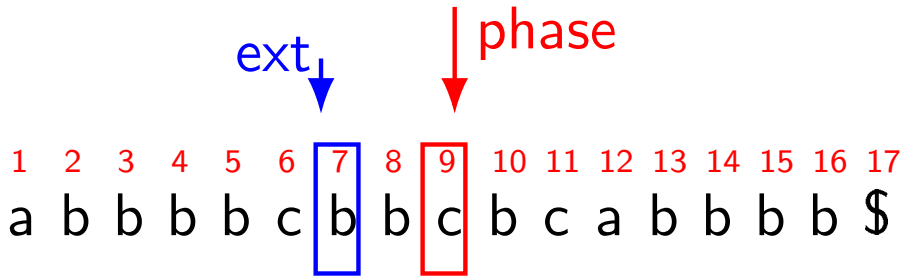
AN :	3 (fllowing the traversal)
rem:	'b'

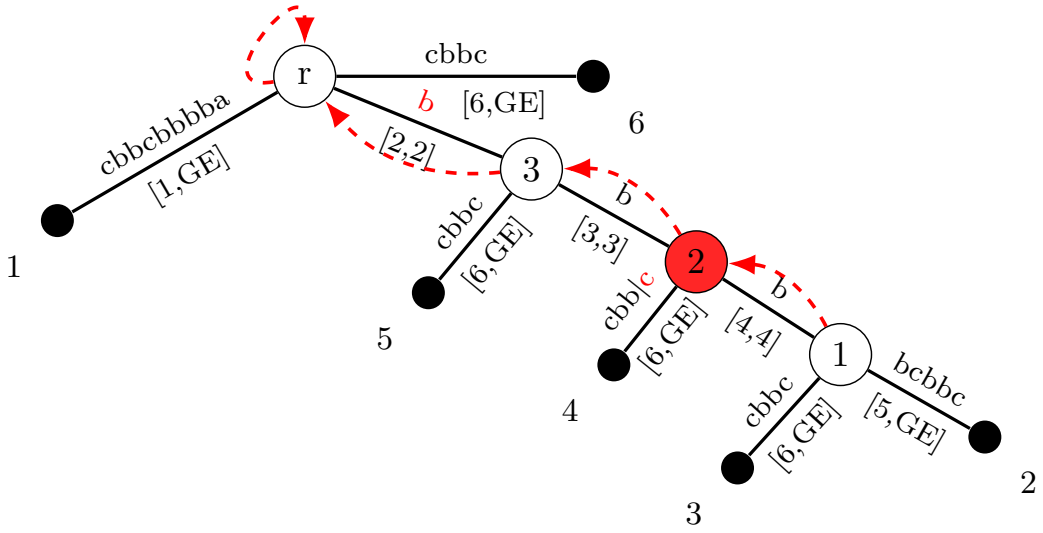
Suffix links

Pending : None

Resolved: None

Phase 9 Extension 7





Phase 9, Extension 7

i :	9
j :	7
GE:	9

Before Extension 7

AN :	3
rem:	'b'

Action

Skip count: skip 'b' from AN

Apply **Rule 3**

After Extension 7

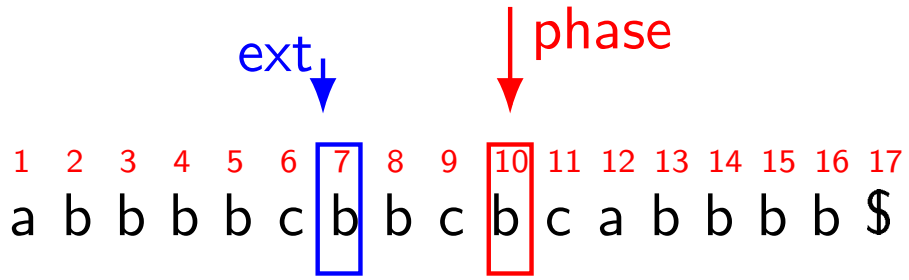
AN :	2 (fllowing the traversal)
rem:	'c'

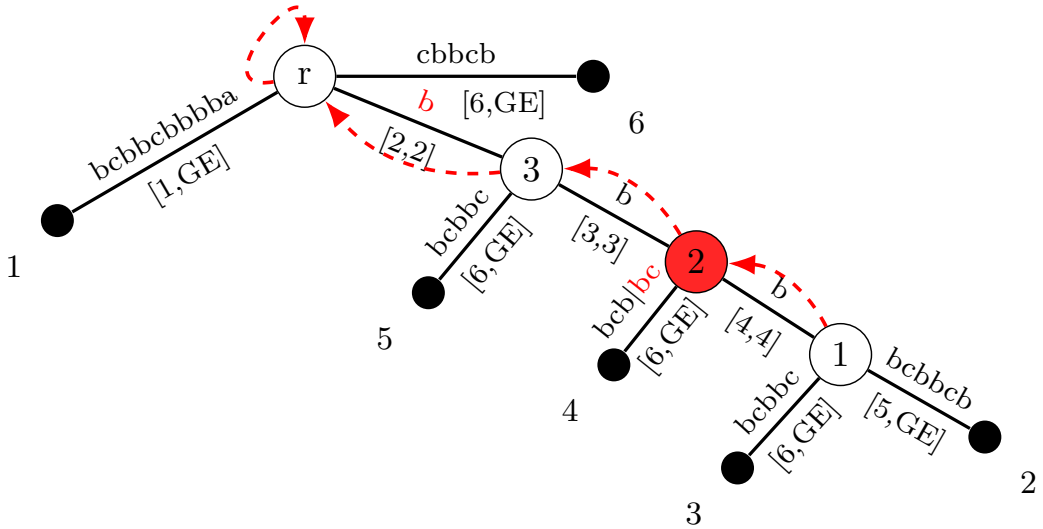
Suffix links

Pending : None

Resolved: None

Phase 10 Extension 7





Phase 10, Extension 7

i :	10
j :	7
GE:	10

Before Extension 7

AN :	2
rem:	'c'

Action

Skip count: skip 'c' from AN

Apply **Rule 3**

After Extension 7

AN :	2 (As previous extension)
rem:	'cb'

Suffix links

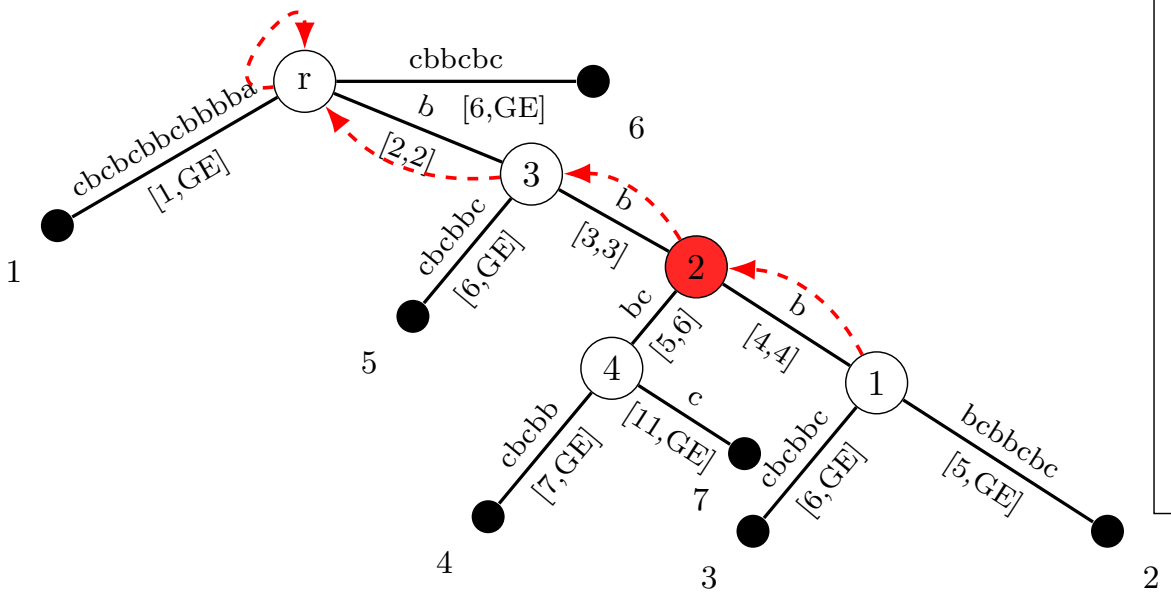
Pending : None

Resolved: None

Phase 11 Extension 7

ext phase

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
a	b	b	b	b	c	b	b	c	b	c	a	b	b	b	b	$



Phase 11, Extension 7

i : 11
j : 7
GE: 11

Before Extension 7

AN : 2
rem: 'cb'

Action

Skip count: skip 'cb' from AN
Apply **Rule 2 Reg**
Internal node created (4)

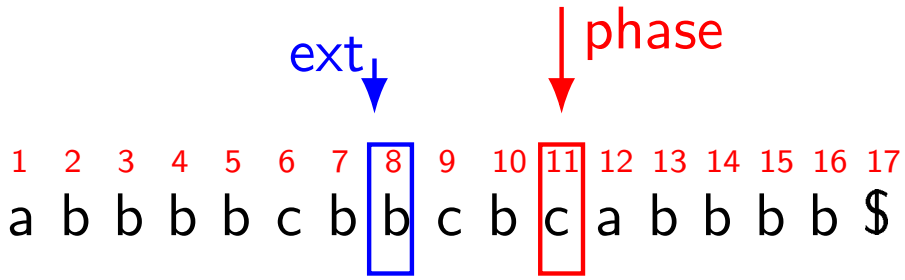
After Extension 7

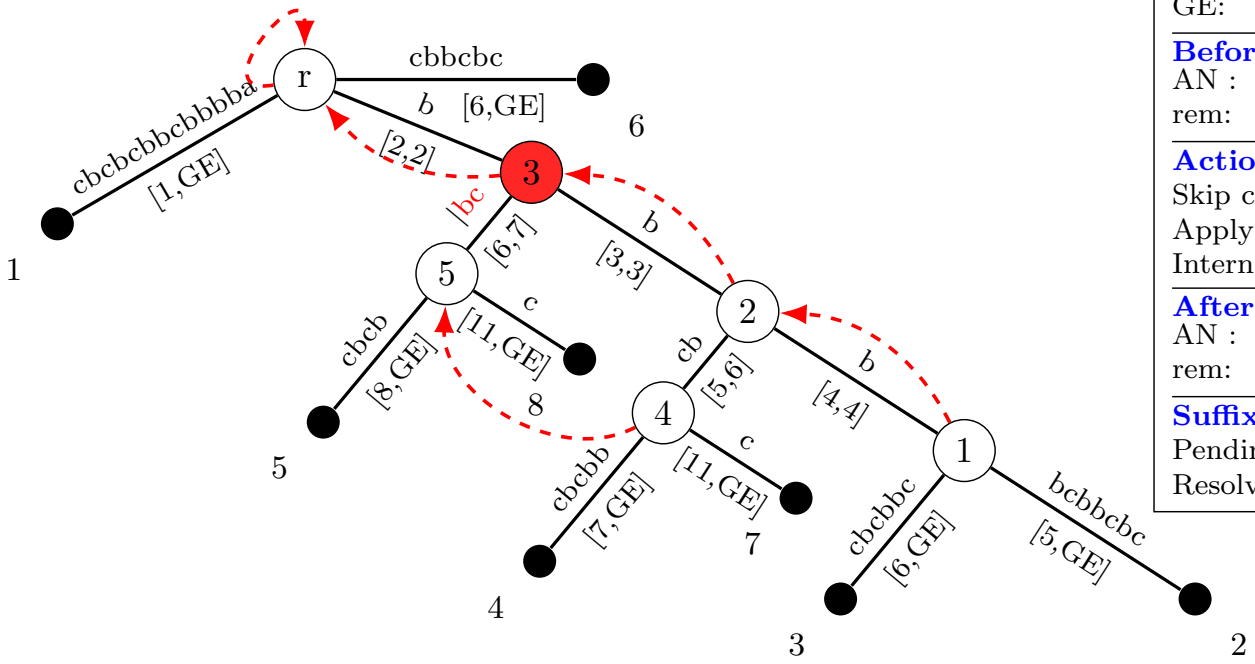
AN : 3 (following link 2 → 3)
rem: 'cb'

Suffix links

Pending : Node 4
Resolved: None

Phase 11 Extension 8





Phase 11, Extension 9

i :	11
j :	8
GE:	11

Before Extension 9

AN :	3
rem:	'cb'

Action

Skip count: skip 'cb' from AN
Apply **Rule 2 Reg**
Internal node created (5)

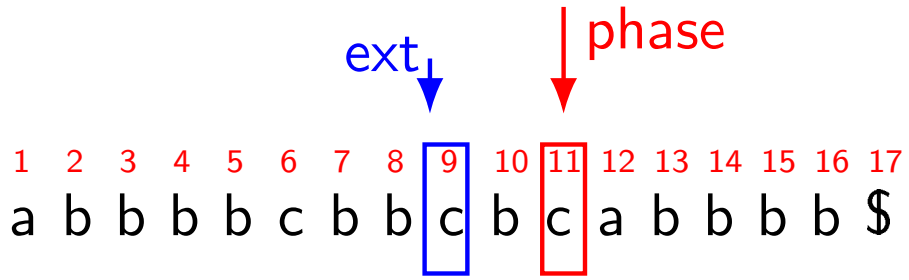
After Extension 9

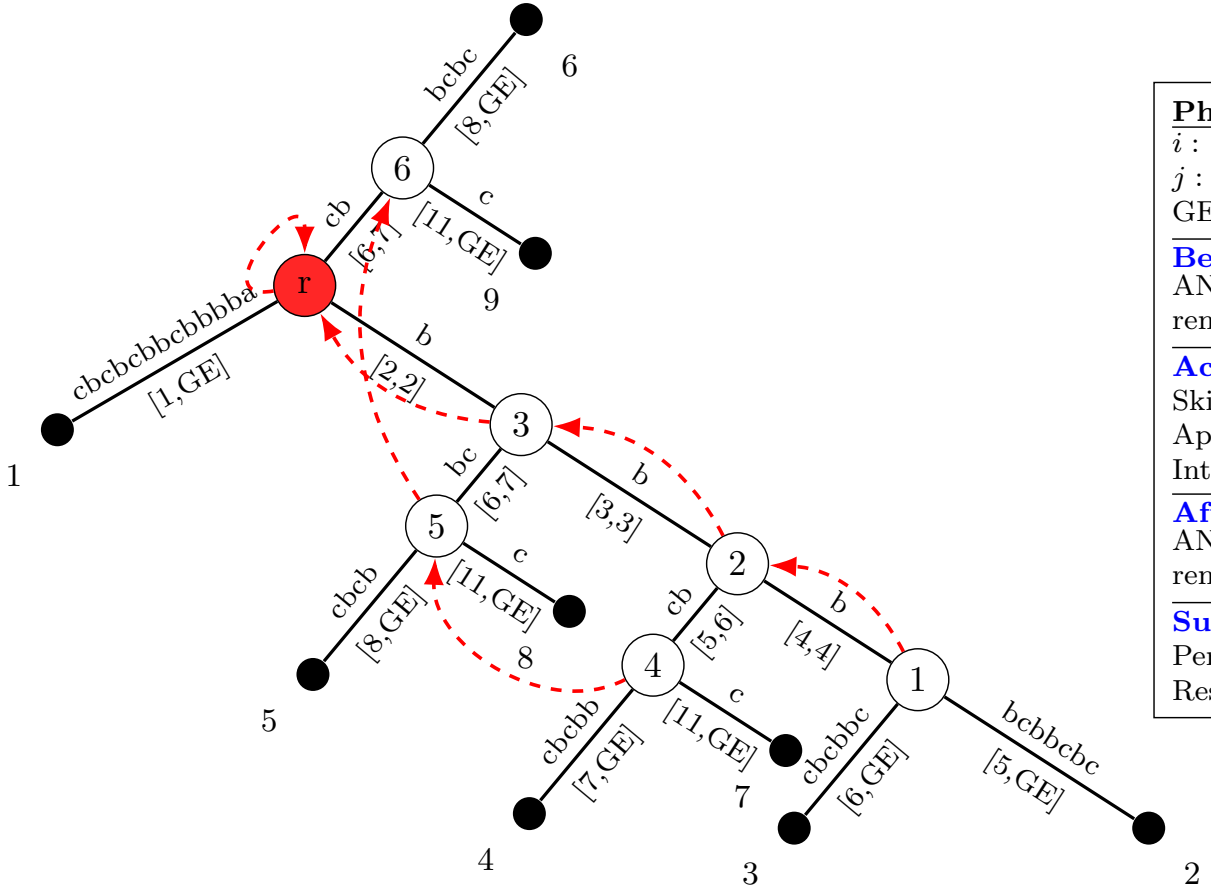
AN :	r (following link 3 → r)
rem:	'cb'

Suffix links

Pending : Node 5
Resolved: 4 → 5

Phase 11 Extension 9





Phase 11, Extension 9

i :	11
j :	9
GE:	11

Before Extension 8

AN :	r
rem:	'cb'

Action

Skip count: skip 'cb' from AN
Apply **Rule 2 Reg**
Internal node created (6)

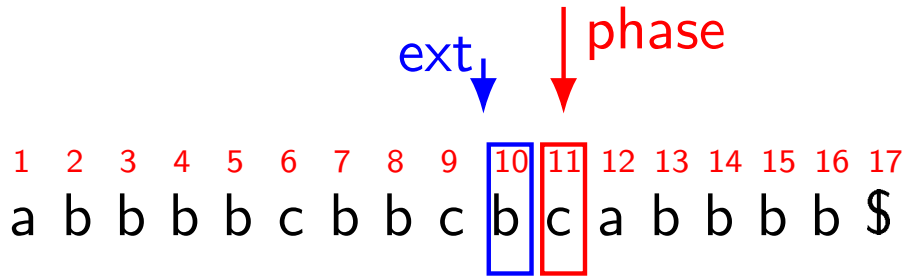
After Extension 8

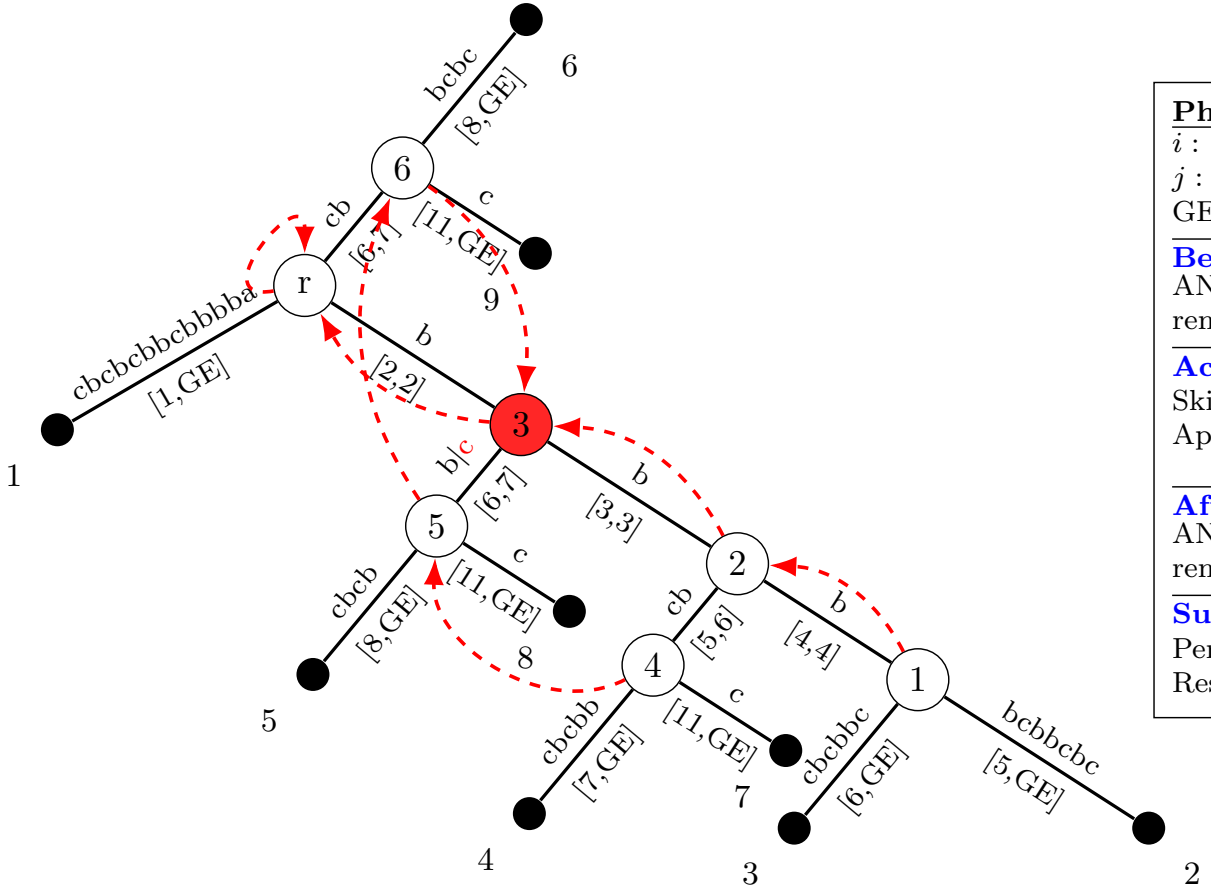
AN :	r (following link 3 → r)
rem:	'b' (remove 1st char)

Suffix links

Pending : Node 6
Resolved: 5 → 6

Phase 11 Extension 10





Phase 11, Extension 10

i :	11
j :	10
GE:	11

Before Extension 10

AN :	r
rem:	'b'

Action

Skip count: skip 'b' from AN

Apply **Rule 3**

After Extension 10

AN :	3 (following skip-count)
rem:	'c'

Suffix links

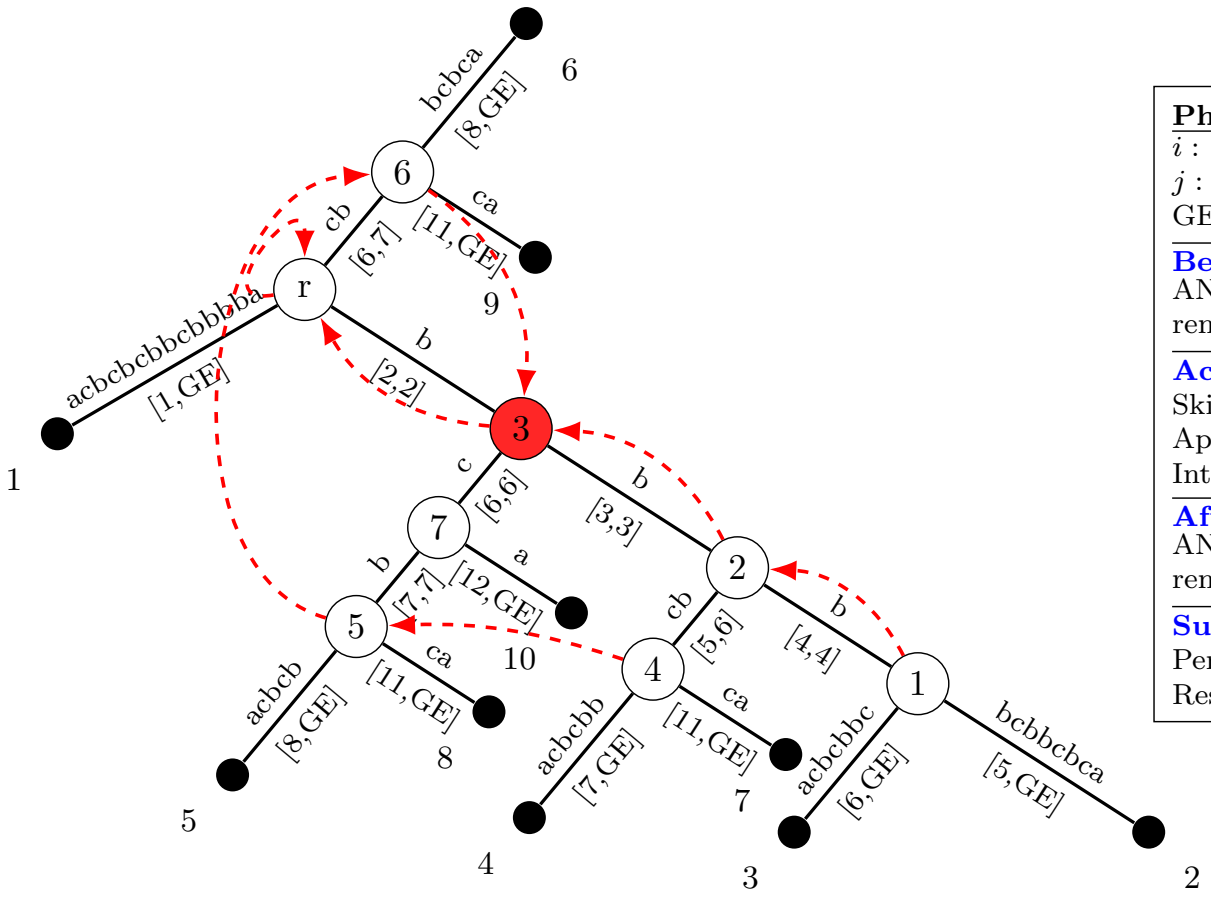
Pending : None

Resolved: 6 → 3 (AN)

Phase 12 Extension 10

ext phase

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
a	b	b	b	b	c	b	b	c	b	c	a	b	b	b	b	$



Phase 12, Extension 10

i : 12
j : 10
GE: 12

Before Extension 10

AN : 3
rem: 'c'

Action

Skip count: skip 'c' from AN
Apply **Rule 2 Reg**
Internal node created (7)

After Extension 10

AN : r (following link 3 → r)
rem: 'c'

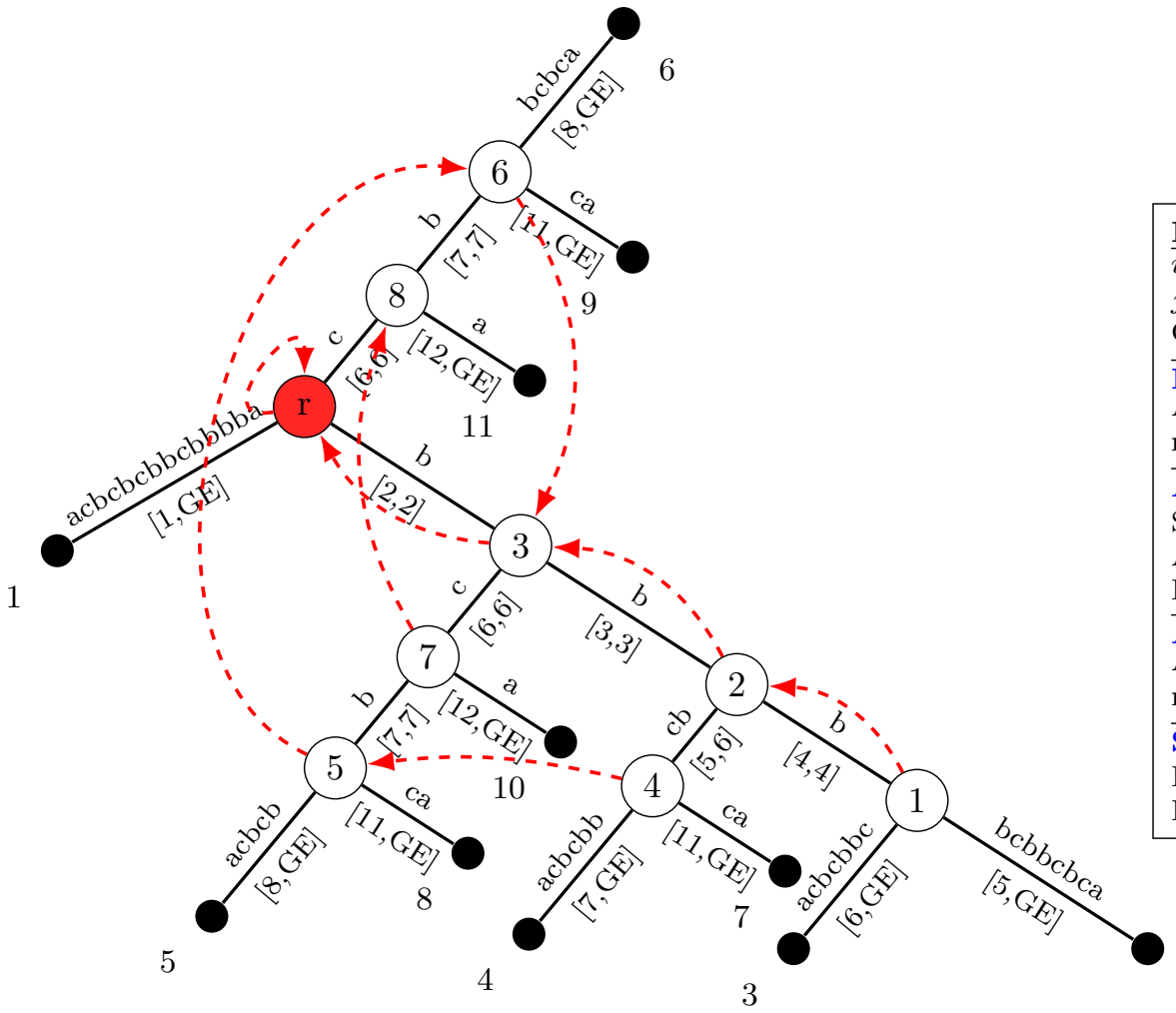
Suffix links

Pending : Node 7
Resolved: None

Phase 12 Extension 11

ext phase

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
a	b	b	b	b	c	b	b	c	b	c	a	b	b	b	b	$



Phase 12, Extension 11

i :	12
j :	11
GE:	12

Before Extension 11

AN :	r
rem:	'c'

Action

Skip count: skip 'c' from AN
Apply **Rule 2 Reg**
Internal node created (8)

After Extension 11

AN :	r (following link r → r)
rem:	' ' (remove 1st char)

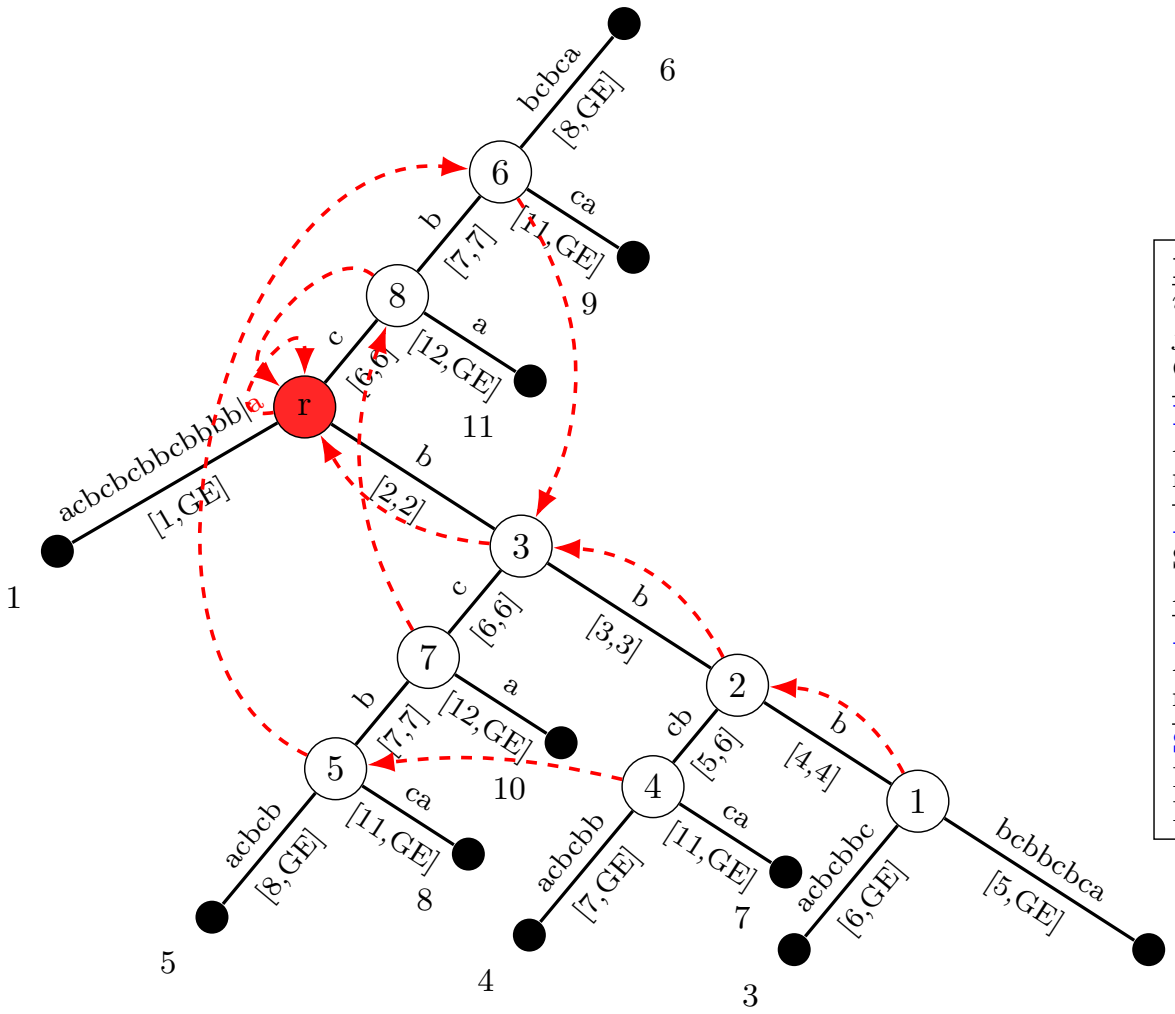
Suffix links

Pending : Node 8
Resolved: 7 → 8

Phase 12 Extension 12

ext phase

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
a	b	b	b	b	c	b	b	c	b	c	a	b	b	b	b	$



Phase 12, Extension 12

i : 12
j : 12
GE: 12

Before Extension 12

AN : r
rem: ‘ ’

Action

Skip count: Not applicable
Apply **Rule 3**

After Extension 12

AN : r (As previous extension)
rem: ‘a’

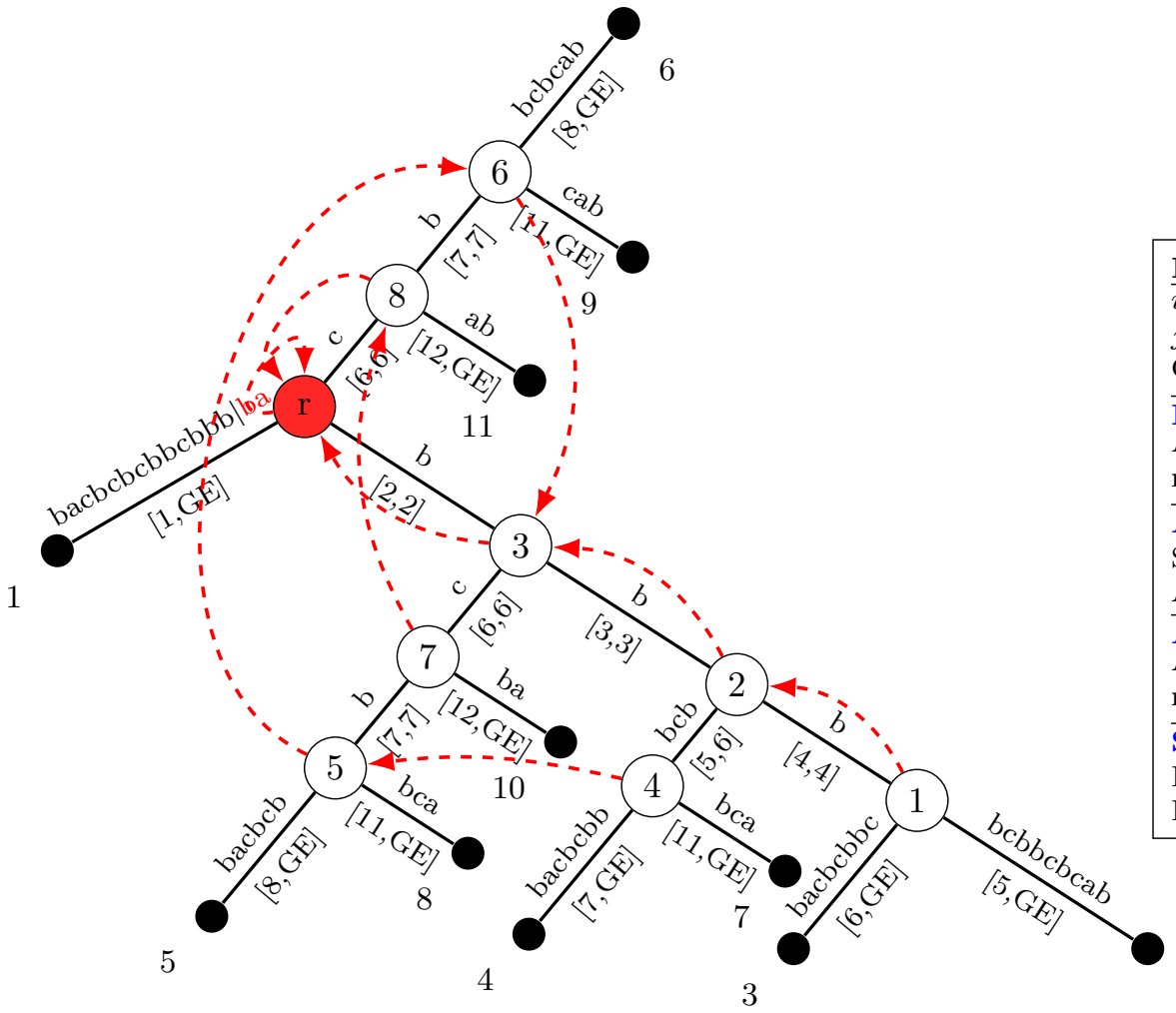
Suffix links

Pending : None
Resolved: 8 → r

Phase 13 Extension 12

ext phase

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
a	b	b	b	b	c	b	b	c	b	c	a	b	b	b	b	$



Phase 13, Extension 12

i :	13
j :	12
GE:	13

Before Extension 12

AN :	r
rem:	'a'

Action

Skip count: skip 'a' from AN

Apply **Rule 3**

After Extension 12

AN :	r (As previous extension)
rem:	'ab'

Suffix links

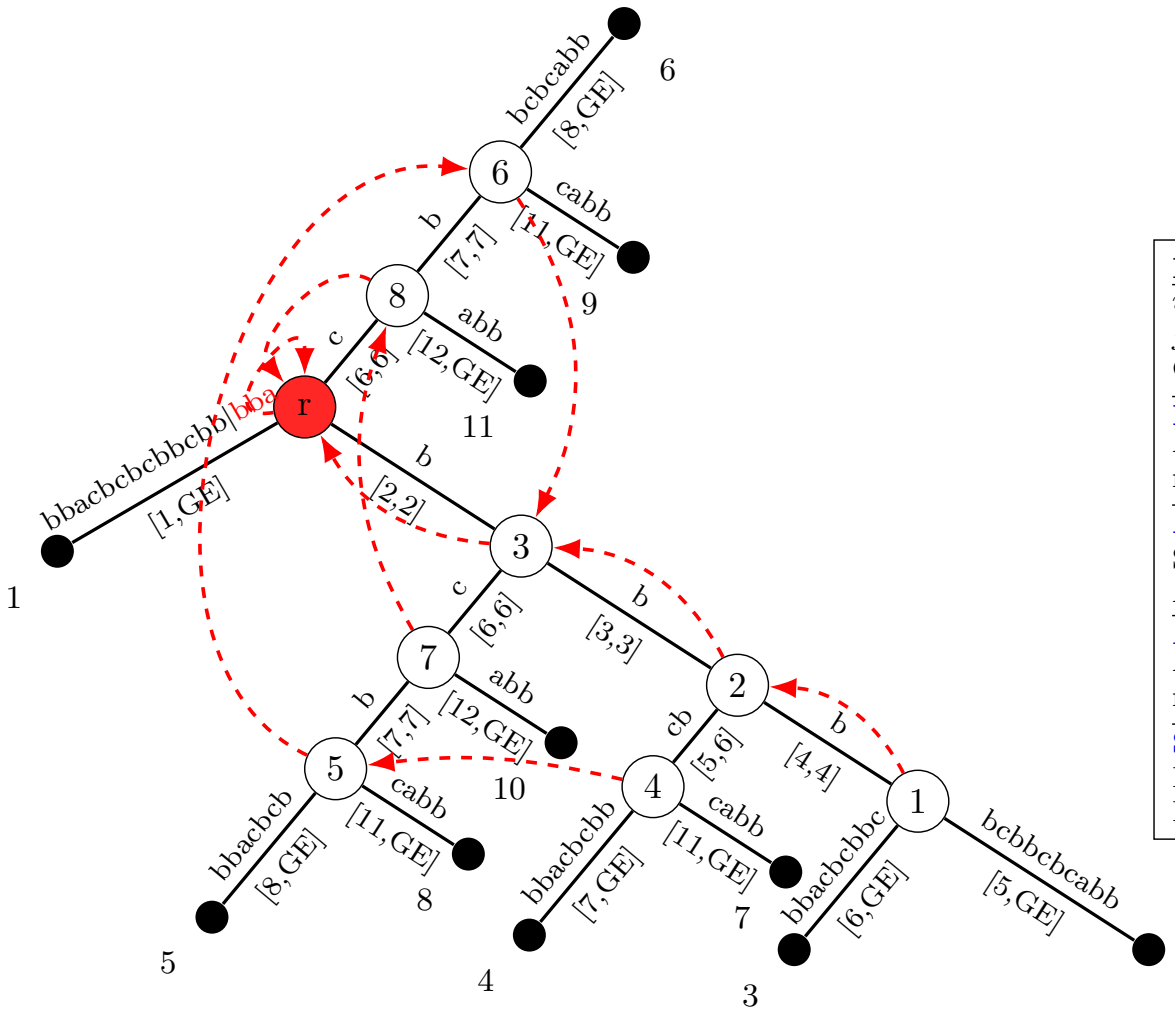
Pending : None

Resolved: None

Phase 14 Extension 12

ext phase

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
a	b	b	b	b	c	b	b	c	b	c	a	b	b	b	b	$



Phase 14, Extension 12

i :	14
j :	12
GE:	14

Before Extension 12

AN :	r
rem:	'ab'

Action

Skip count: skip 'ab' from AN

Apply **Rule 3**

After Extension 12

AN :	r (As previous extension)
rem:	'abb'

Suffix links

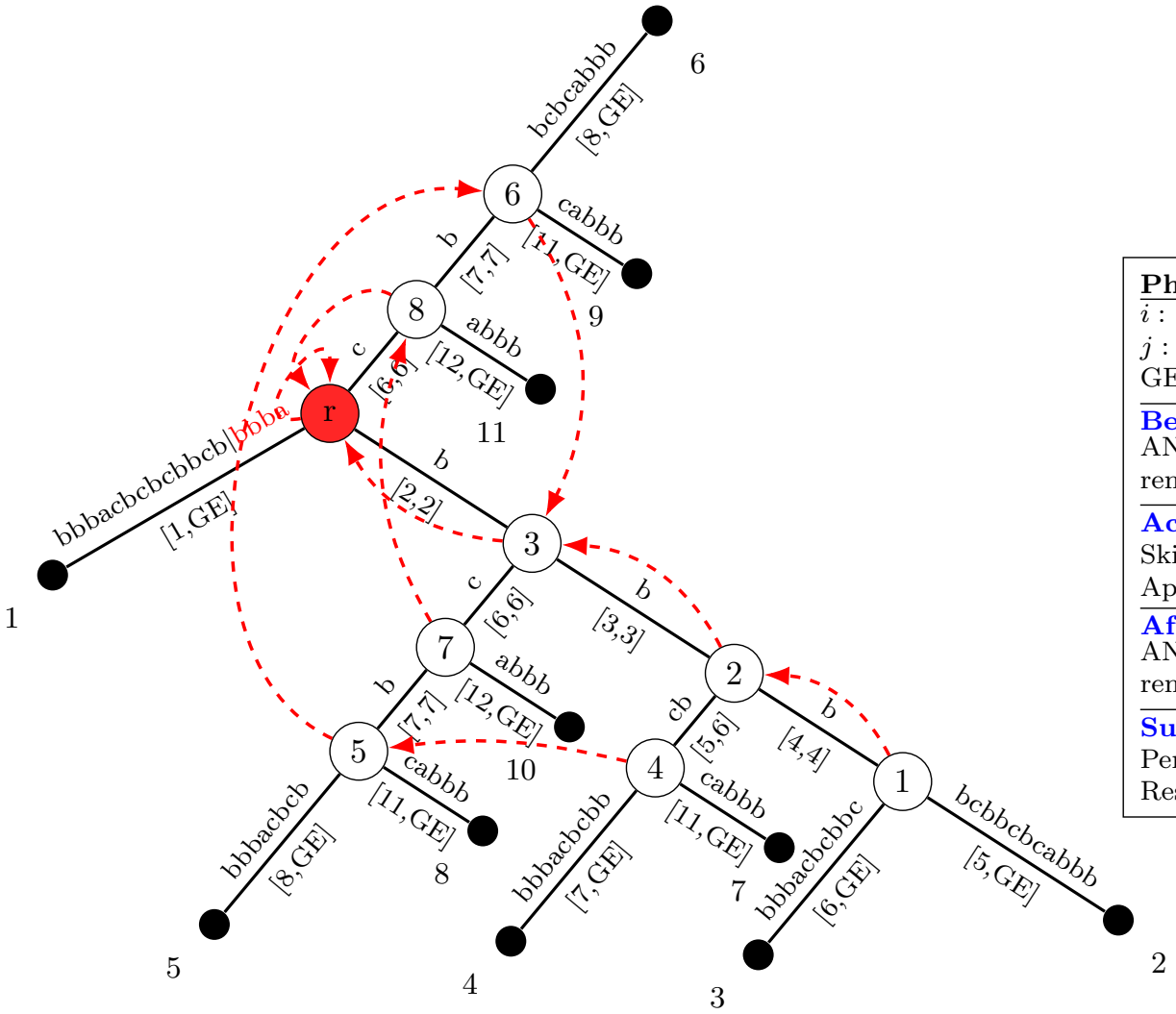
Pending : None

Resolved: None

Phase 15 Extension 12

ext phase

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
a	b	b	b	b	c	b	b	c	b	c	a	b	b	b	b	$



Phase 15, Extension 12

i :	15
j :	12
GE:	15

Before Extension 12

AN :	r
rem:	'abb'

Action

Skip count: skip 'abb' from AN

Apply **Rule 3**

After Extension 12

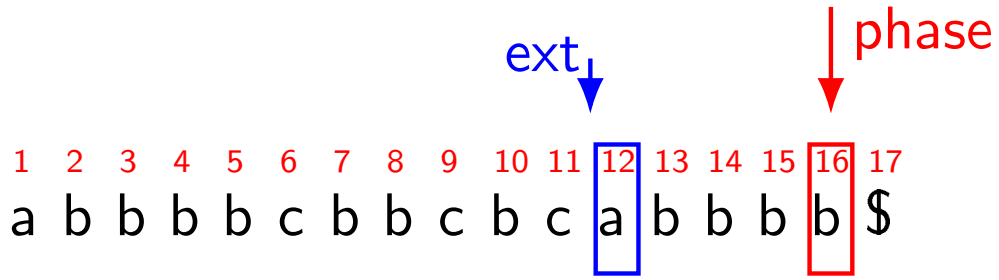
AN :	r (As previous extension)
rem:	'abbb'

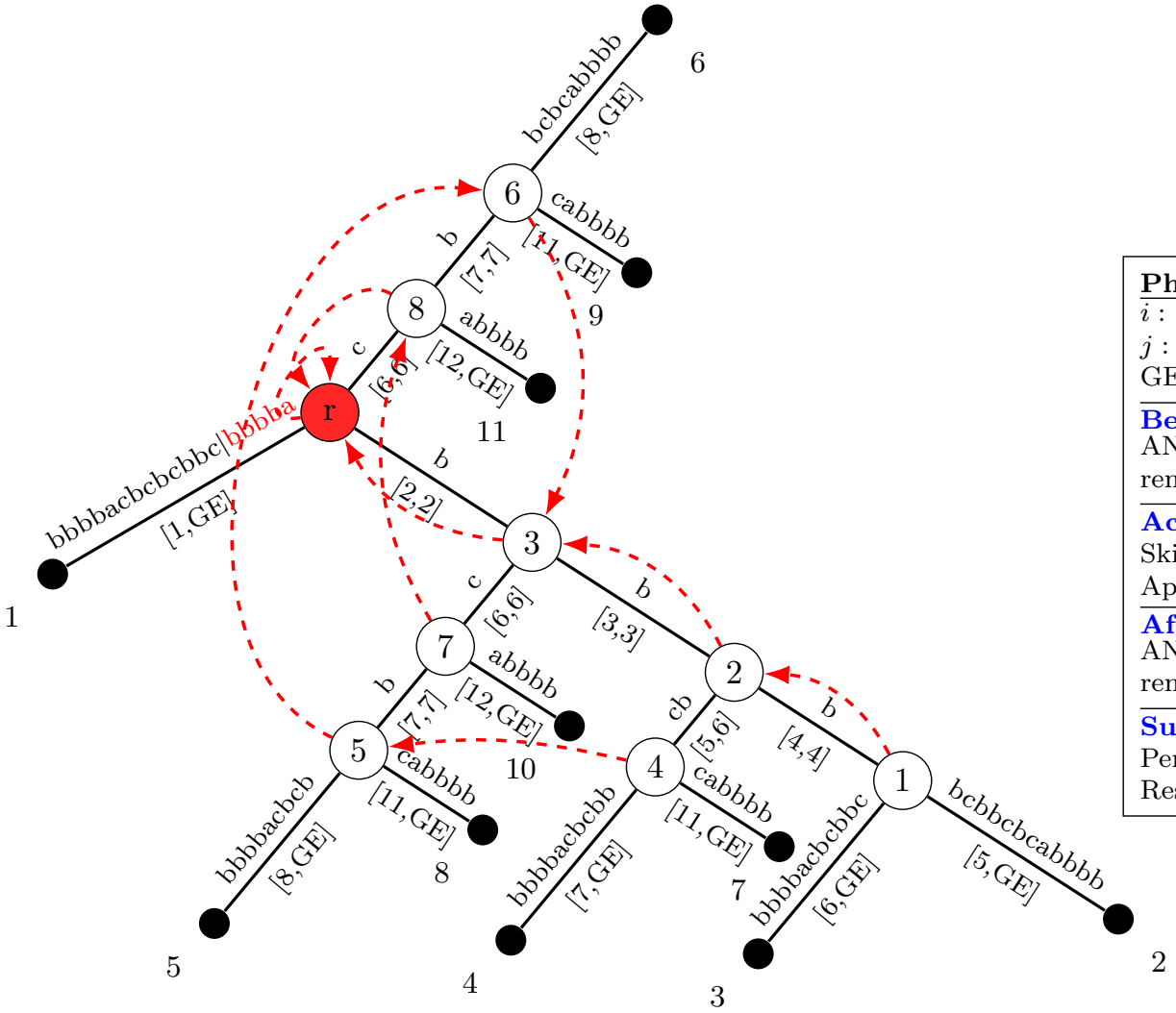
Suffix links

Pending : None

Resolved: None

Phase 16 Extension 12





Phase 16, Extension 12

i :	16
j :	12
GE:	16

Before Extension 12

AN :	r
rem:	'abbb'

Action

Skip count: skip 'abbb' from AN

Apply **Rule 3**

After Extension 12

AN :	r (As previous extension)
rem:	'abbbb'

Suffix links

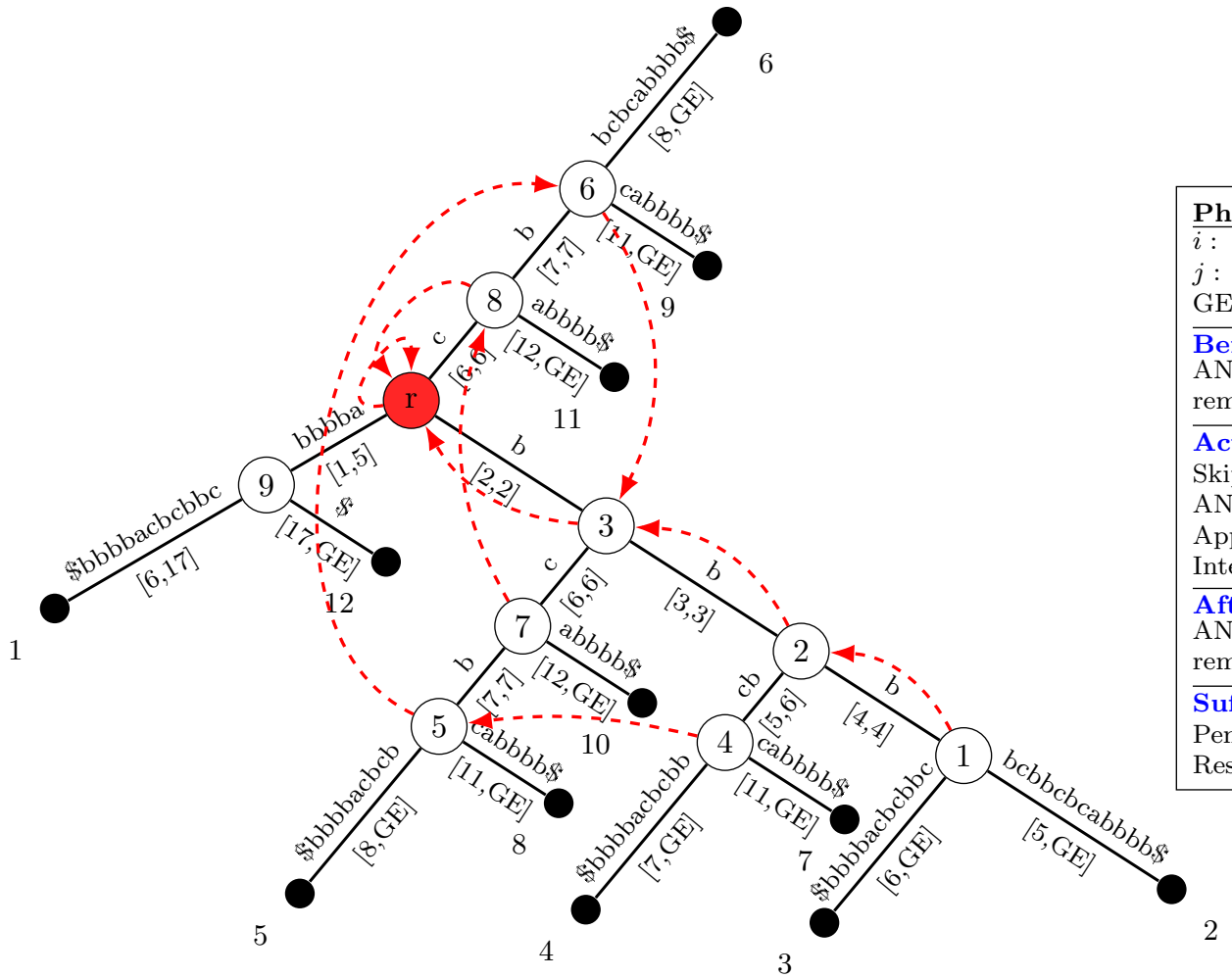
Pending : None

Resolved: None

Phase 17 Extension 12

ext phase

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
a	b	b	b	b	c	b	b	c	b	c	a	b	b	b	b	$



Phase 17, Extension 12

i :	17
j :	12
GE:	17

Before Extension 12

AN : r
rem: 'abbbb'

Action

Skip count: skip 'abbbb' from AN
Apply **Rule 2 Reg**
Internal node created (9)

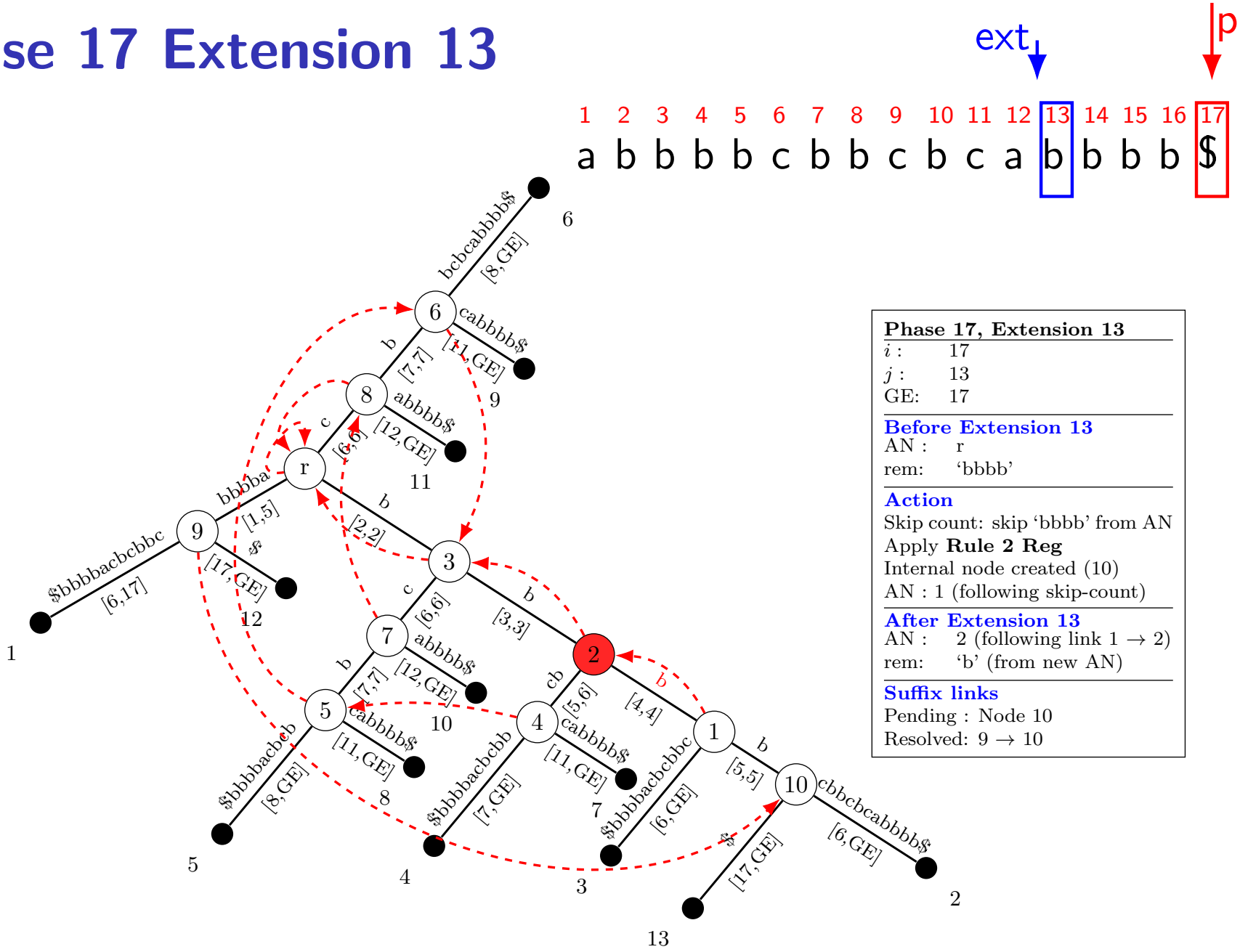
After Extension 12

AN : r (following link r → r)
rem: 'bbbb' (remove 1st char)

Suffix links

Pending : Node 9
Resolved: None

Phase 17 Extension 13



Phase 17, Extension 13

i :	17
j :	13
GE:	17

Before Extension 13

AN :	r
rem:	'bbbb'

Action

Skip count: skip 'bbbb' from AN
Apply **Rule 2 Reg**
Internal node created (10)
AN : 1 (following skip-count)

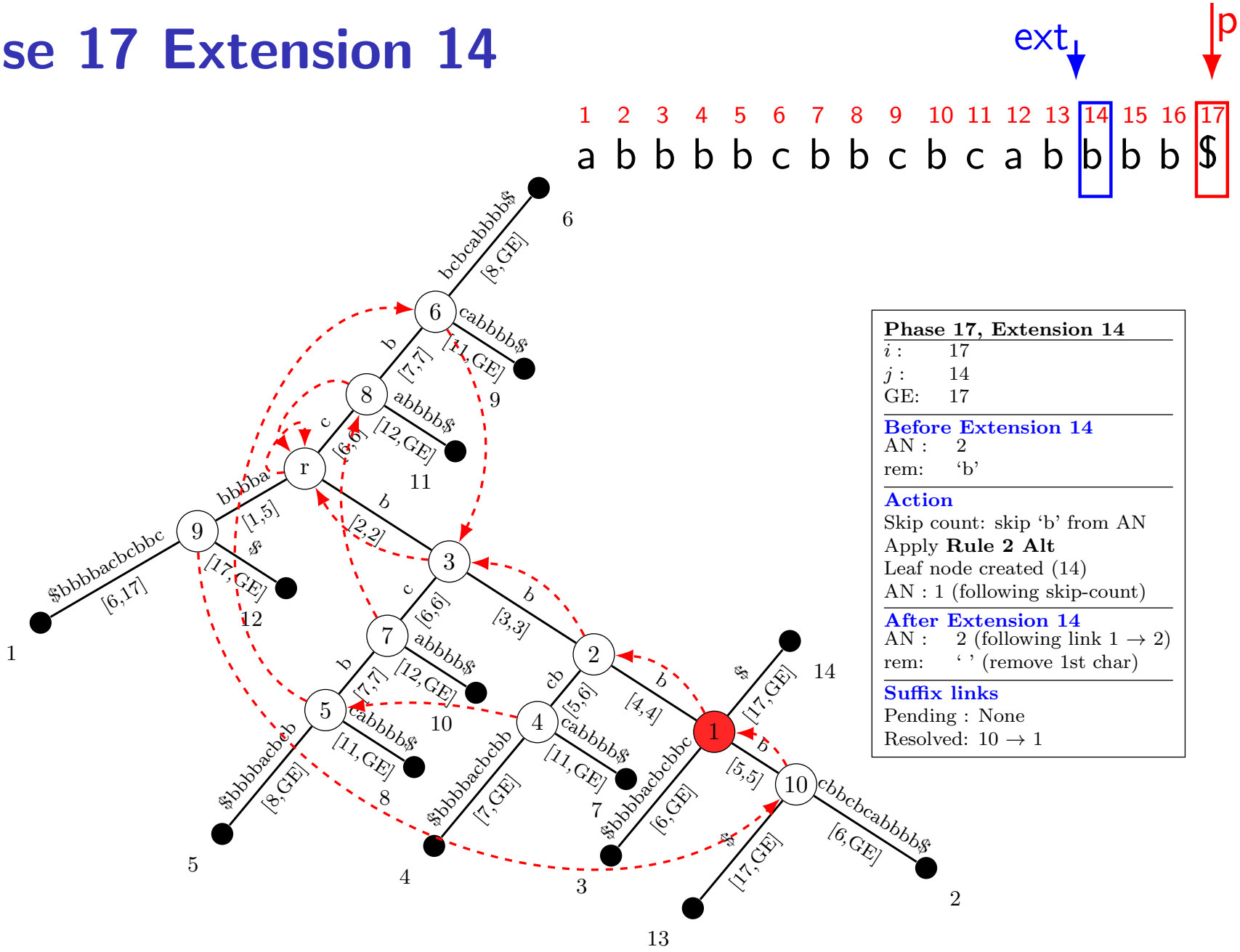
After Extension 13

AN :	2 (following link 1 → 2)
rem:	'b' (from new AN)

Suffix links

Pending : Node 10
Resolved: 9 → 10

Phase 17 Extension 14



Phase 17, Extension 14

i :	17
j :	14
GE:	17

Before Extension 14

AN :	2
rem:	'b'

Action

Skip count: skip 'b' from AN
Apply **Rule 2 Alt**
Leaf node created (14)
AN : 1 (following skip-count)

After Extension 14

AN :	2 (following link 1 → 2)
rem:	' ' (remove 1st char)

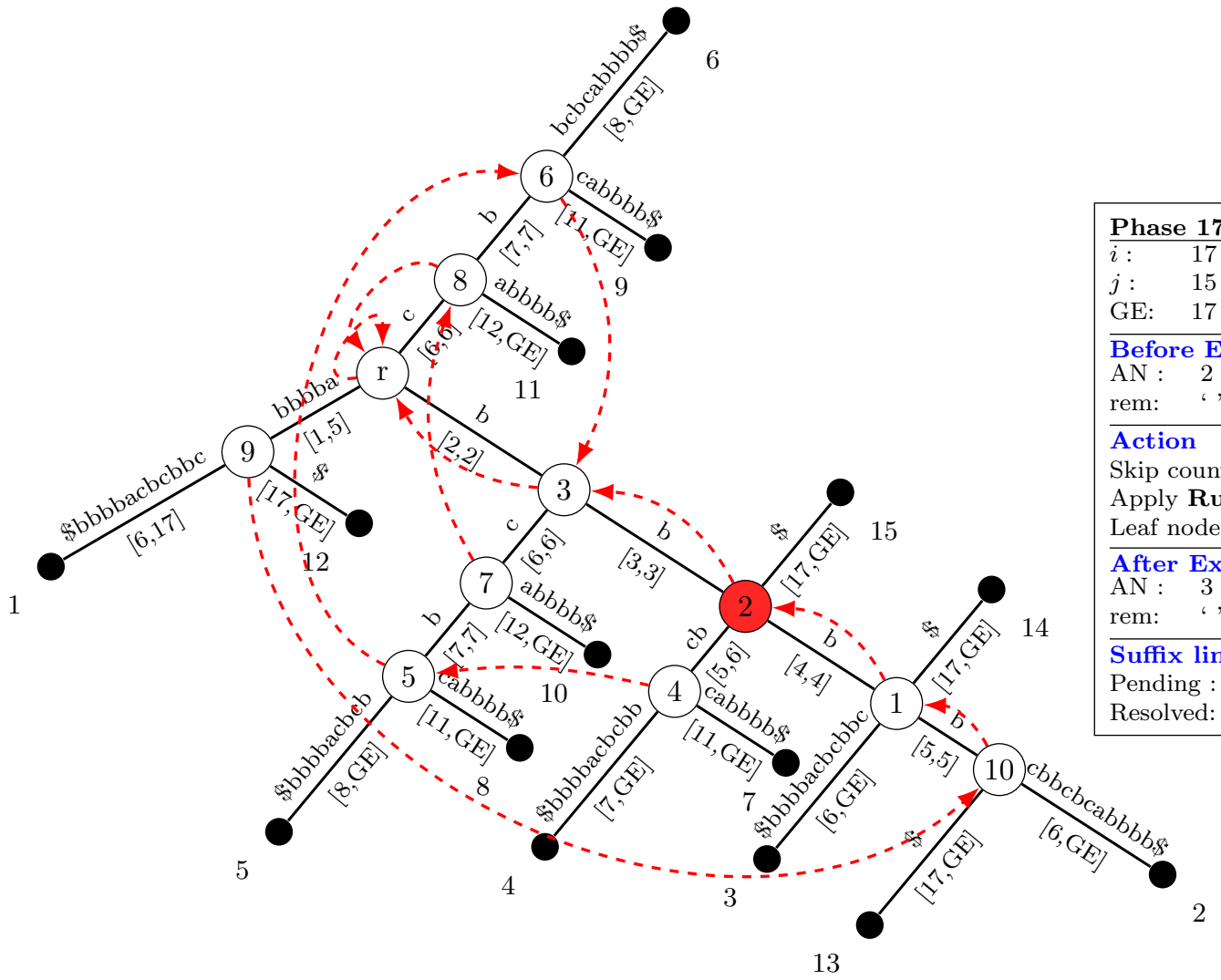
Suffix links

Pending : None
Resolved: 10 → 1

Phase 17 Extension 15

ext phase

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
a	b	b	b	b	c	b	b	c	b	c	a	b	b	b	b	$



Phase 17, Extension 15

i :	17
j :	15
GE:	17

Before Extension 15

AN :	2
rem:	‘ ’

Action

Skip count: not applicable
Apply **Rule 2 Alt**
Leaf node created (15)

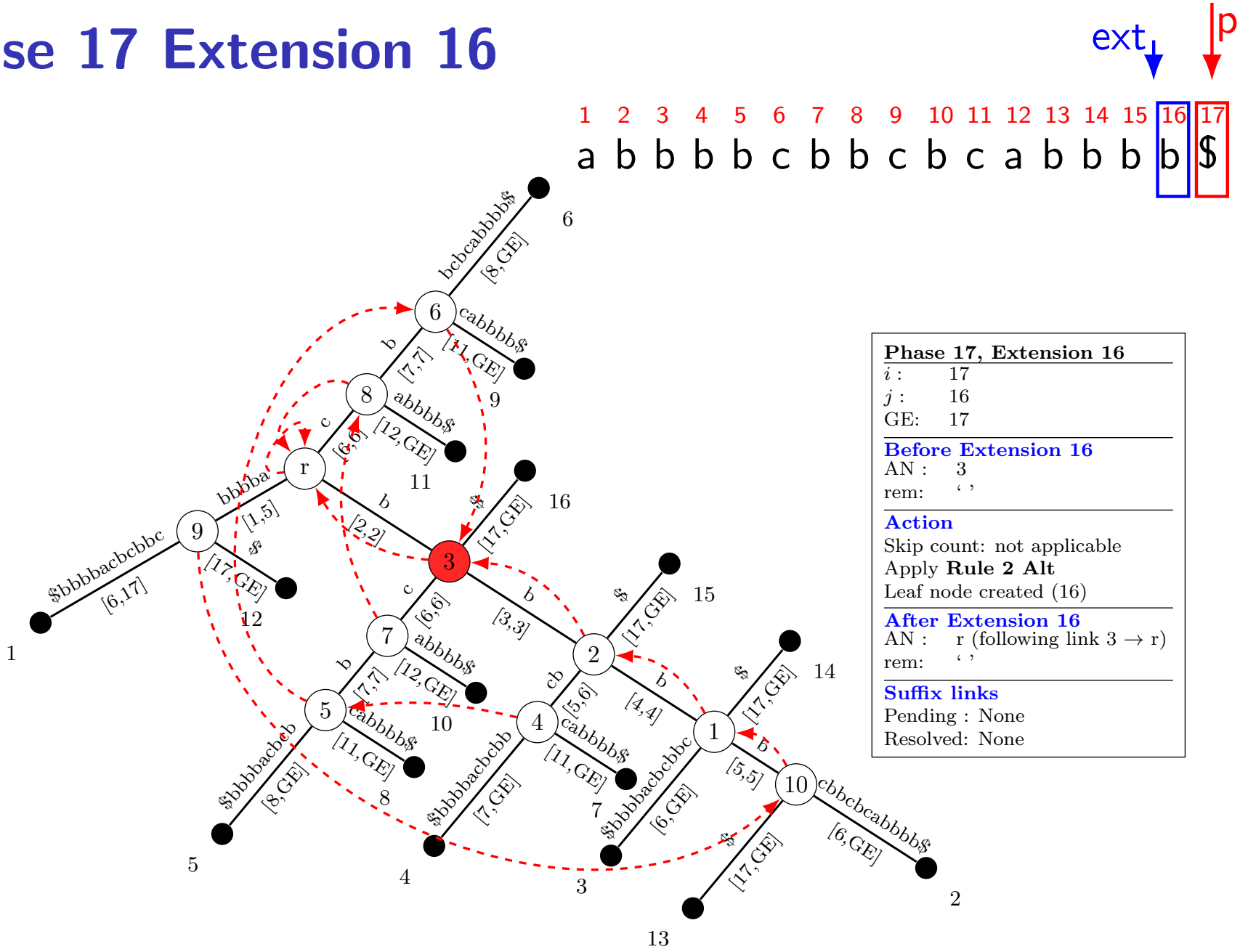
After Extension 15

AN :	3 (following link 2 → 3)
rem:	‘ ’

Suffix links

Pending : None
Resolved: None

Phase 17 Extension 16



Phase 17, Extension 16

i :	17
j :	16
GE:	17

Before Extension 16

AN :	3
rem:	‘ ’

Action

Skip count: not applicable
Apply **Rule 2 Alt**
Leaf node created (16)

After Extension 16

AN :	r (following link 3 → r)
rem:	‘ ’

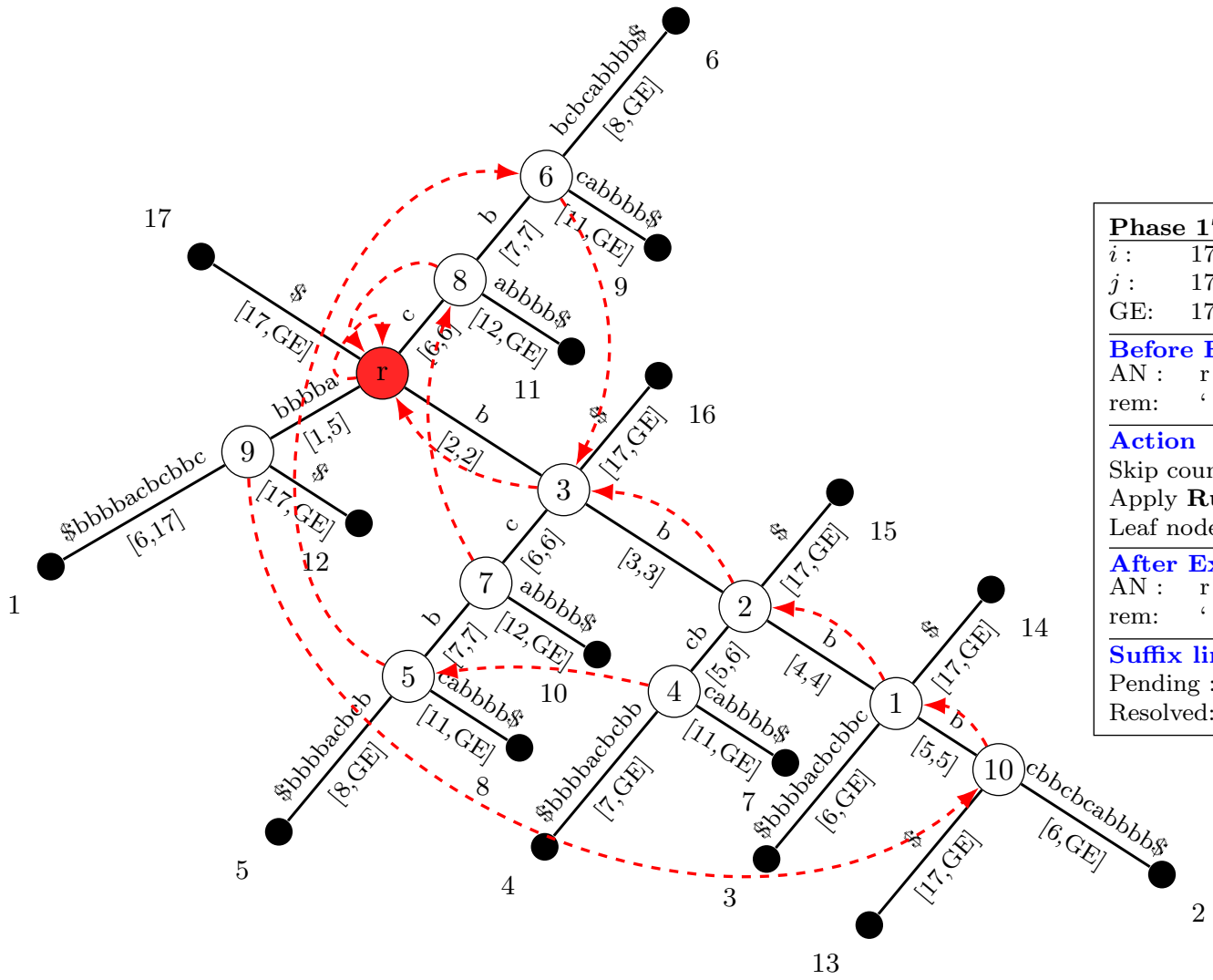
Suffix links

Pending : None
Resolved: None

Phase 17 Extension 17

ext phase

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
a	b	b	b	b	c	b	b	c	b	c	a	b	b	b	b	$



Phase 17, Extension 17

i :	17
j :	17
GE:	17

Before Extension 17

AN :	r
rem:	‘ ’

Action

Skip count: not applicable
Apply **Rule 2 Alt**
Leaf node created (17)

After Extension 17

AN :	r (following link r → r)
rem:	‘ ’

Suffix links

Pending : None
Resolved: None